

Polynomial Regression: Turbine Score and Thermal Anomaly

Approach, degree choices and what I checked along the way

1. The data and the plan

Each problem comes with 1000 training rows and 1000 test rows, no missing values and no duplicate rows. All inputs lie in $[-1, 1]$, and a large share of them sit exactly on -1 or $+1$: about 30% of the problem 1 training values and about 24% of the problem 2 ones. So the inputs were clipped at some point, which matters later. I used degree in the assignment's sense, the sum of the powers in each term is at most d . That gives 83 terms at degree 3 for six features and 454 terms at degree 12 for three features, so the number of terms passes the 800 rows I train on in each CV fold quite quickly. I therefore judged every choice by cross-validated MSE (5 folds, repeated several times for the final calls) and never by training error. I also kept a 20% hold-out, redoing the whole selection on the other 80%, as a separate sanity check.

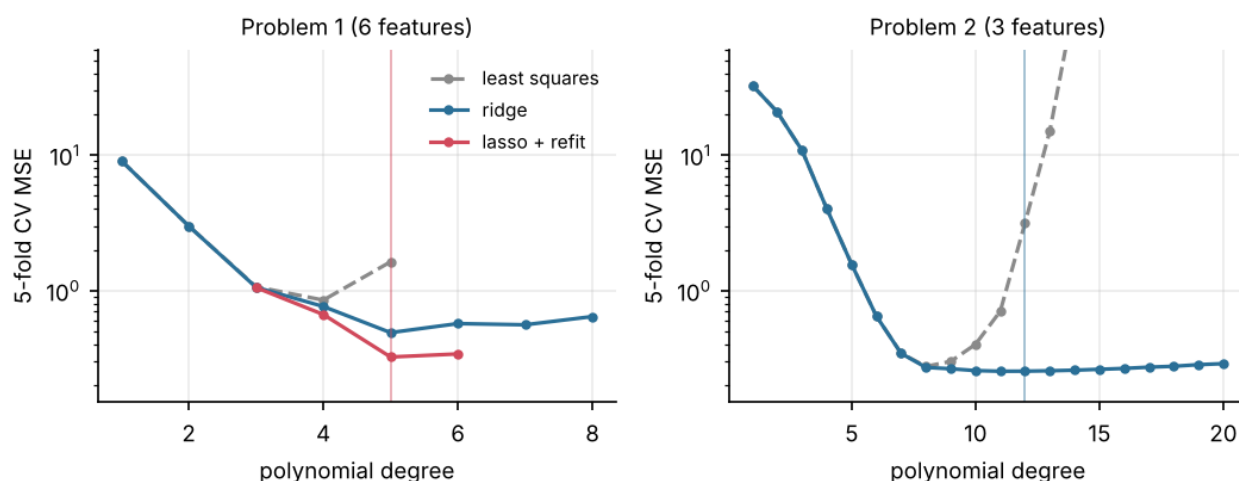


Figure 1. 5-fold CV error against degree. Plain least squares blows up as soon as the term count nears the sample size; the penalised fits stay stable. Vertical lines mark the degrees I used.

2. Problem 1: turbine net power score

Degree 5, with the terms chosen by lasso and then refit by ordinary least squares. Error falls steadily up to degree 5 and then stops improving. Ridge on the full set of 461 degree-5 terms gets about 0.49, but a lasso on the same terms gets about 0.33, which told me the true function is sparse: only some of the 461 terms matter, and ridge wastes effort shrinking all of them. Lasso alone shrinks the surviving coefficients too, so I use it only to choose terms and then refit those by plain least squares on the unshrunk data. The penalty is picked by 5-fold CV repeated three times. The final model keeps 70 terms (CV MSE 0.323). Going to degree 6 or 7 only added candidate terms and made the CV error slightly worse (0.34 and 0.36 with the same method), so I stayed at 5.

degree	terms	least squares	ridge	lasso + refit
3	83	1.069	1.061	1.055
4	209	0.852	0.765	0.668
5	461	1.634	0.491	0.324
6	923	-	0.572	0.342
7	1715	-	0.560	-
8	3002	-	0.645	-

Table 1. Problem 1, 5-fold CV MSE (best penalty for each cell). A dash means not run, or more terms than rows.

The test inputs are not distributed like the training inputs. In the test file about half of all values sit at ± 1 , against about 30% in training, so test rows live much nearer the corners of the cube (Figure 2, left). Only 3 training rows have all six inputs clipped, and 15 have five. That is why my test predictions have a standard deviation of about 4.3 while the training targets have 3.2. The spread grows with the number of clipped inputs in the same way in both files, so it is the inputs rather than the model that explain it. I did not want the shift to flip my choice of model, so I re-scored the candidates giving each validation row a weight equal to the test share divided by the training share of rows with the same number of clipped inputs. Lasso + refit at degree 5 stayed best: 0.352 weighted CV MSE (0.327 unweighted). Degree 4 gave 0.90, degree 6 gave 0.38, and the best ridge

fits were 0.54 or worse. Refitting with those weights as well was worse (0.39), so the submitted model is fitted without weights. My honest expectation for the test MSE is therefore around 0.35 to 0.4 rather than the 0.32 that plain CV reports.

3. Problem 2: thermal anomaly score

Degree 12 with ridge penalty 0.2. Here the function is dense rather than sparse: lasso never beat ridge at any degree I tried (0.27 at degree 10, against 0.25 for ridge). Error drops quickly until degree 8, where plain least squares is still fine (0.276 with 164 terms). After that least squares degrades rapidly (0.40 at degree 10, 3.2 at degree 12) because the system is nearly singular, while ridge keeps improving a little and then flattens between degrees 10 and 13. To decide inside that flat stretch I ran 10 repeats of 5-fold CV and compared the options on identical splits. Degree 12 with penalty 0.2 had the lowest mean error (0.2526); degrees 10, 11 and 13 were each behind by 0.003 to 0.005, which is small but was consistent across repeats (paired standard errors 0.0008 to 0.0014). Beyond degree 13 the error creeps up again, to 0.29 at degree 20, so the 'up to degree 20' in the problem statement does not mean I needed to use it.

degree	terms	least squares	ridge
6	83	0.650	0.650
8	164	0.276	0.272
10	285	0.402	0.257
11	363	0.706	0.255
12	454	3.163	0.255
14	679	136.4	0.259
16	968	-	0.267
20	1770	-	0.290

Table 2. Problem 2, 5-fold CV MSE from the single-pass sweep. The final degree was settled by the repeated-CV comparison above (degree 12, 0.2526).

In problem 2 the test inputs look like the training inputs (the shares of rows with 0, 1, 2 or 3 clipped inputs match within about 4 percentage points), so I used ordinary CV there without any reweighting.

4. Underfitting and overfitting

Problem 1	degree	terms used	train MSE	CV MSE
	2	23	2.85	3.00
	3	54	0.87	1.06
	4	73	0.50	0.67
	5 (used)	72	0.25	0.33
	6	100	0.24	0.34
Problem 2	degree	terms used	train MSE	CV MSE
	2	9	20.10	20.86
	4	34	3.59	4.03
	6	83	0.47	0.65
	8	164	0.18	0.27
	10	285	0.17	0.26
	12 (used)	454	0.17	0.25
	16	968	0.16	0.27
	20	1770	0.16	0.29

Table 3. Error on the training data against 5-fold CV error, each degree fitted with its own best penalty. Same recipe as the final model for each problem.

Underfitting is the low-degree end of both tables: train and CV error are both large and close to each other (problem 1 at degree 2: 2.85 and 3.00; problem 2 at degree 4: 3.59 and 4.03), which means the polynomial simply cannot bend enough. **Overfitting** is the other end. In problem 2 the training error keeps creeping down as the degree grows (0.170 at degree 12, 0.158 at degree 20) while the CV error turns around and gets worse (0.254 to 0.290), and unpenalised least squares falls apart completely (Figure 1). That turning point is why I stopped at degree 12 and why I used penalties. I am not claiming the final models have zero overfitting: both still

have a gap of about 0.08 between train and CV error (0.25 against 0.33, and 0.17 against 0.25). With hundreds of coefficients fitted on 1000 noisy rows some of that gap is unavoidable, and the hold-out error matched the CV error, so I read it as small and under control.

5. Hold-out check and final numbers

	model	CV MSE	hold-out MSE	hold-out R2
Problem 1	degree 5, 70 terms	0.323	0.326	0.964
Problem 2	degree 12, ridge 0.2	0.253	0.201	0.995

Table 4. The hold-out is 200 rows that were not used for choosing the degree, the penalty or the coefficients.

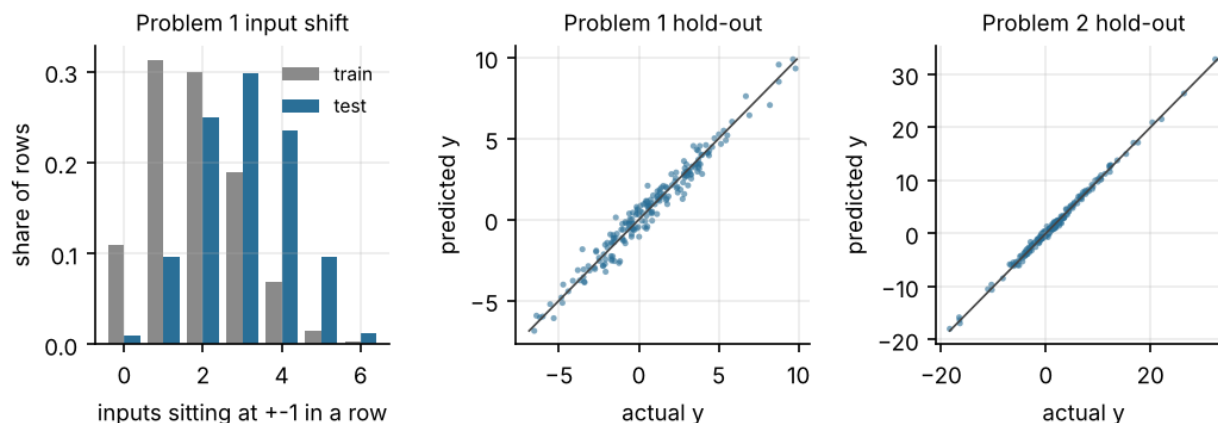


Figure 2. Left: share of rows by how many of the six inputs sit at +-1, train against test. Middle and right: hold-out predictions against the true values.

The hold-out numbers match the CV numbers closely, which is what I wanted to see. The hold-out for problem 2 came out a bit better than CV (0.20 against 0.25), which I read as ordinary luck on 200 rows rather than a sign that CV is pessimistic, so I would quote 0.25 as my expectation there. The final models were refit on all 1000 rows. The prediction files have one column, y , 1000 rows each, in the order of the test files, matching the sample submission.

6. Other things I tried, and what they cost me

Plain least squares at every degree (fails once the number of terms nears the sample size). Backward elimination by t-statistic on the degree-5 terms (0.49 at the one threshold I let finish, clearly worse than lasso and very slow). Orthogonal matching pursuit with 50 to 90 terms (best 0.35 at 60 terms, close but behind lasso + refit at 0.32). A Legendre-polynomial basis with ridge for problem 2 (0.36 at degree 10 and worse above that). It spans the same polynomials but the penalty shrinks the wrong things, and the plain monomial basis was much better. Importance-weighted fitting for problem 1 (0.39, covered above).

7. Limits I want to be upfront about

I do not have the test targets, so every figure above is from the training data. For problem 1 in particular the test rows sit in a region where I have little training data, so the real test error could be higher than my estimate if the true function behaves differently near the corners. The degree-5 structure with 70 active terms fits the data well, but I cannot prove that it is the exact generating function; I only know it generalises better than the alternatives I compared. Every model is a polynomial in the inputs fitted by least squares, but problem 1 uses lasso to choose its terms and both use a ridge or lasso penalty, so these are penalised polynomial regressions rather than plain unpenalised ones. All code is in the repository, and the numbers in this report come from running it; the side experiments in section 6 are in the experiments folder.